INCH-POUND
MIL-DTL-24643/13F
27 November 2023
SUPERSEDING
MIL-DTL-24643/13E
1 October 2009

DETAIL SPECIFICATION SHEET

# CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TYPE LSTTRS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

## Construction (Non-Watertight)

| | | |
|---|---|---|
| First | - | Copper conductor, coated or uncoated, 20 AWG, class B stranding in accordance with ASTM B8. |
| Second | - | Separator may be used at manufacturer's option where required to provide free-stripping insulation. |
| Third | - | Thermoset insulation. Nominal wall thickness of 0.020 inch; wall thickness may vary from nominal as necessary in order to meet the specified electrical requirements. |
| Fourth | - | Two conductors, one black and one white (or natural), shall be cabled together to form a pair. The length of lay shall be determined to permit a completed cable to meet the twisting and bending endurance tests. |
| Fifth | - | An optional binder. |
| Sixth | - | Braided shield of 34 or 36 AWG, coated or uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent. |
| Seventh | - | Shield insulation of two polyester tapes with the outer tape sealed. An alternate shield insulation may consist of one polyester tape plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. Standard identification code shall be applied by method 2. |
| Eighth | - | The specified number of pairs (see table I) shall be cabled together. The length of lay(s) shall be required for the completed cable to meet the twisting and bending endurance tests. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. A single pair shall not be used as center for the cable core. Outer layer shall be left hand. Inner layer(s) direction of lay shall be at the manufacturer's option. Fillers may be used as necessary to form a firm, well-rounded assembly. |
| Ninth | - | An optional binder. |
| Tenth | - | Cross-linked polyolefin jacket. See table I for thickness. |



Distribution Statement A. Approved for public release: distribution is unlimited.

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NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details.

| Military part no. M24643/13 | Type and size | Number of pairs | Jacket wall thickness min. avg. (inch) | Overall diameter (inches) | | Roller diameter max. (inches) |
|---|---|---|---|---|---|---|
| | | | | min. | max. | |
| -01UN | LSTTRS-2 | 2 | 0.050 | 0.629 | 0.680 | 2.0 |
| -02UN | LSTTRS-4 | 4 | 0.050 | 0.685 | 0.740 | 2.2 |
| -03UN | LSTTRS-6 | 6 | 0.050 | 0.814 | 0.880 | 3.0 |
| -04UN | LSTTRS-8 | 8 | 0.050 | 0.916 | 0.990 | 3.5 |
| -05UN | LSTTRS-10 | 10 | 0.050 | 0.999 | 1.080 | 4.0 |
| -06UN | LSTTRS-12 | 12 | 0.070 | 1.017 | 1.100 | 4.0 |
| -07UN | LSTTRS-16 | 16 | 0.070 | 1.100 | 1.190 | 4.5 |

REQUIREMENTS:

Qualification required.

INSPECTION:

**Basic electricals:**

| | |
|---|---|
| Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.) | 11.95 |
| Voltage withstand (volts, root mean square, min.) | |
| Conductor to conductor | 2,000 |
| Conductor to shield | 1,000 |
| Component shield to component shield | 200 |
| Insulation resistance (megohms/1,000 feet, min.) | |
| Conductor to conductor | 500 |
| Conductor to shield | 200 |
| Component shield to component shield | 100 |
| Conductor continuity | No failure |
| Shield continuity | No failure |

**Group A:**

| | |
|---|---|
| Visual and dimensional | No failure |
| Capacitance | |
| Mutual capacitance (nominal at 1 megahertz [MHz]) (picofarad [pF]/100 feet) | 28 |
| Capacitance unbalance (percent, max.) | 10 |



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| | |
|---|---|
| Characteristic impedance – at 1 MHz (ohms) | |
| Minimum | 75 |
| Maximum | 90 |
| **Group B:** | |
| Thermoset proof test (percent, max.) | |
| Insulation | 50 |
| Jacket (when tested at 392 °F [200 °C]) | 50 |
| Twisting and bending endurance (see table I for roller size) | |
| At 167 °F (75 °C) (cycles, min.) | 6,000 |
| At -22±3.6 °F (-30±2 °C) (cycles, min.) | 2,000 |
| Physicals (unaged) | |
| Insulation (extruded insulation only) | |
| Tensile strength (lb/in$^2$, min.) | 700 |
| Elongation (percent, min.) | 150 |
| Jacket (cable) | |
| Tensile strength (lb/in$^2$, min.) | 1,300 |
| Elongation (percent, min.) | 160 |
| Tear (lb/in thickness, min.) | 35 |
| **Group C:** | |
| Physicals (aged) (air oven) | |
| Insulation (extruded insulation only) | |
| Tensile strength (percent of unaged, min.) | 75 |
| Elongation (percent of unaged, min.) | 75 |
| Jacket (cable) | |
| Tensile strength (percent of unaged, min.) | 60 |
| Elongation (percent of unaged, min.) | 60 |
| Permanence of printing (jacket) (cycles, min.) | 125 |
| Heat distortion (percent of unaged, max.) | 30 |
| Shrinkage | No failure |
| Shield (conformance to material, construction, and coverage) | No failure |



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**Group D:**

| | |
|---|---|
| Flame propagation (cable) | No failure |

**Qualification inspection:**

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

| | |
|---|---|
| Aging and compatibility (cable) (257±9 °F [125±5 °C]) | No failure |
| Abrasion resistance (jacket) (scrapes, min.) | 75 |
| Acid gas equivalent (percent, max.) | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Shield insulation | 2 |
| Halogen content (percent, max.) | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Insulation | 0.2 |
| Immersion (jacket) | |
| Tensile strength (percent of unaged, min.) | 50 |
| Elongation (percent of unaged, min.) | 50 |
| Smoke index (max.) | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 45 |
| Toxicity index (max.) | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) – type A (hardness, min.) | 80 |
| Weathering (jacket) | No failure |
| Electrical moisture absorption | No failure |

UNIT ORDERING LENGTHS:

| Type and size | Feet (nominal) |
|---|---|
| LSTTRS all sizes | 1,500 |



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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-029)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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